

3633. Earliest Finish Time for Land and Water Rides I

Difficulty: Easy

Link: <https://leetcode.com/problems/earliest-finish-time-for-land-and-water-rides-i/?envType=daily-question&envId=2026-06-02>

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Problem:

You are given two categories of theme park attractions: **land rides** and **water rides**.

- **Land rides**

- `landStartTime[i]` – the earliest time the i^{th} land ride can be boarded.
- `landDuration[i]` – how long the i^{th} land ride lasts.

- **Water rides**

- `waterStartTime[j]` – the earliest time the j^{th} water ride can be boarded.
- `waterDuration[j]` – how long the j^{th} water ride lasts.

A tourist must experience **exactly one** ride from **each** category, in **either order**.

- A ride may be started at its opening time or **any later moment**.
- If a ride is started at time t , it finishes at time $t + \text{duration}$.
- Immediately after finishing one ride the tourist may board the other (if it is already open) or wait until it opens.

Return the **earliest possible time** at which the tourist can finish both rides.

Example 1:

Input: landStartTime = [2,8], landDuration = [4,1], waterStartTime = [6], waterDuration = [3]

Output: 9

Explanation:

- Plan A (land ride 0 → water ride 0):
 - Start land ride 0 at time $\text{landStartTime}[0] = 2$. Finish at $2 + \text{landDuration}[0] = 6$.
 - Water ride 0 opens at time $\text{waterStartTime}[0] = 6$. Start immediately at 6 , finish at $6 + \text{waterDuration}[0] = 9$.
- Plan B (water ride 0 → land ride 1):
 - Start water ride 0 at time $\text{waterStartTime}[0] = 6$. Finish at $6 + \text{waterDuration}[0] = 9$.
 - Land ride 1 opens at $\text{landStartTime}[1] = 8$. Start at time 9 , finish at $9 + \text{landDuration}[1] = 10$.
- Plan C (land ride 1 → water ride 0):
 - Start land ride 1 at time $\text{landStartTime}[1] = 8$. Finish at $8 + \text{landDuration}[1] = 9$.
 - Water ride 0 opened at $\text{waterStartTime}[0] = 6$. Start at time 9 , finish at $9 + \text{waterDuration}[0] = 12$.
- Plan D (water ride 0 → land ride 0):
 - Start water ride 0 at time $\text{waterStartTime}[0] = 6$. Finish at $6 + \text{waterDuration}[0] = 9$.
 - Land ride 0 opened at $\text{landStartTime}[0] = 2$. Start at time 9 , finish at $9 + \text{landDuration}[0] = 13$.

Plan A gives the earliest finish time of 9.

Example 2:

Input: landStartTime = [5], landDuration = [3], waterStartTime = [1], waterDuration = [10]

Output: 14

Explanation:

- Plan A (water ride 0 → land ride 0):

- Start water ride 0 at time `waterStartTime[0] = 1` . Finish at `1 + waterDuration[0] = 11` .
- Land ride 0 opened at `landStartTime[0] = 5` . Start immediately at `11` and finish at `11 + landDuration[0] = 14` .

- Plan B (land ride 0 → water ride 0):

- Start land ride 0 at time `landStartTime[0] = 5` . Finish at `5 + landDuration[0] = 8` .
- Water ride 0 opened at `waterStartTime[0] = 1` . Start immediately at `8` and finish at `8 + waterDuration[0] = 18` .

Plan A provides the earliest finish time of 14.

Constraints:

- `1 <= n, m <= 100`
- `landStartTime.length == landDuration.length == n`
- `waterStartTime.length == waterDuration.length == m`
- `1 <= landStartTime[i], landDuration[i], waterStartTime[j], waterDuration[j] <= 1000`